

Concurrency Control

Some Slides taken from Mohammad Hammoud (CMU, Qatar).

Schedules

- A **schedule** is a list of actions (i.e., read, write, abort, and/or commit) from a *set* of transactions
- The *order* in which two actions of a transaction **T** appear in a schedule must be the same as they appear in **T** itself
- Assume **T1** = [R(A), W(A)] and **T2** = [R(B), W(B), R(C), W(C)]

T1	T2
R(A)	
	R(B)
W(A)	
	W(B)
	R(C)
	W(C)



T1	T2
R(A)	
W(A)	
	R(B)
	W(B)
	R(C)
	W(C)



T1	T2
R(A)	
W(A)	
	R(C)
	W(C)
	R(B)
	W(B)



Serial Schedules

- A **complete schedule** must contain all the actions of every transaction that appears on it
- If the actions of different transactions are not interleaved, the schedule is called a **serial schedule**

T1	T2
R(A)	
W(A)	
Commit	
	R(B)
	W(B)
	R(C)
	W(C)
	Commit

A Serial Schedule

T1	T2
	R(B)
	W(B)
R(A)	
W(A)	
Commit	
	R(C)
	W(C)
	Commit

A Non-Serial Schedule

Anomalies

- Interleaving actions of different transactions can leave the database in an **inconsistent state**
- Two actions on the same data object are said to ***conflict*** if at least one of them is a write
- There are 3 **anomalies** that can arise upon interleaving actions of different transactions (say, T1 and T2):
 - **Write-Read (WR) Conflict**: T2 reads a data object previously written by T1
 - **Read-Write (RW) Conflict**: T2 writes a data object previously read by T1
 - **Write-Write (WW) Conflict**: T2 writes a data object previously written by T1

Implementation of Isolation Levels

- Locking
 - Lock on whole database vs lock on items
 - How long to hold lock?
 - Shared vs exclusive locks
- Timestamps
 - Transaction timestamp assigned e.g. when a transaction begins
 - Data items store two timestamps
 - Read timestamp
 - Write timestamp
 - Timestamps are used to detect out of order accesses
- Multiple versions of each data item
 - Allow transactions to read from a “snapshot” of the database



Lock-Based Protocols

- WR, RW and WW anomalies can be avoided using a *locking protocol*
- A lock is a mechanism to control concurrent access to a data item
- Data items can be locked in two modes :
 1. **exclusive** (*X*) *mode*. Data item can be both read as well as write. X-lock is requested using **lock-X** instruction.
 2. **shared** (*S*) *mode*. Data item can only be read. S-lock is requested using **lock-S** instruction.
- Lock requests are made to concurrency-control manager. Transaction can proceed only after request is granted.
- The part of the DBMS that keeps track of locks is called the *lock manager*



Lock-Based Protocols (Cont.)

- **Lock-compatibility matrix**

	S	X
S	true	false
X	false	false

- A transaction may be granted a lock on an item if the requested lock is compatible with locks already held on the item by other transactions
- Any number of transactions can hold shared locks on an item,
- But if any transaction holds an exclusive on the item, no other transaction may hold any lock on the item.



Lock-Based Protocols (Cont.)

- Example of a transaction performing locking:

T_2 : **lock-S**(A);

read (A);

unlock(A);

lock-S(B);

read (B);

unlock(B);

display($A+B$)

- Locking as above is not sufficient to guarantee serializability



Schedule With Lock Grants

- Grants omitted in rest of chapter
 - Assume grant happens just before the next instruction following lock request
- This schedule is not serializable (why?)
- A **locking protocol** is a set of rules followed by all transactions while requesting and releasing locks.
- Locking protocols enforce serializability by restricting the set of possible schedules.

T_1	T_2	concurrency-control manager
lock-X(B)		grant-X(B, T_1)
read(B)		
$B := B - 50$		
write(B)		
unlock(B)		
	lock-S(A)	grant-S(A, T_2)
	read(A)	
	unlock(A)	
	lock-S(B)	grant-S(B, T_2)
	read(B)	
	unlock(B)	
	display($A + B$)	
lock-X(A)		grant-X(A, T_1)
read(A)		
$A := A + 50$		
write(A)		
unlock(A)		



Deadlock

- Locking can lead to an undesirable situation. For example, consider the partial schedule

T_3	T_4
lock-X(B)	
read(B)	
$B := B - 50$	
write(B)	
	lock-S(A)
	read(A)
	lock-S(B)
lock-X(A)	

- Neither T_3 nor T_4 can make progress — executing **lock-S(B)** causes T_4 to wait for T_3 to release its lock on B , while executing **lock-X(A)** causes T_3 to wait for T_4 to release its lock on A .
- Such a situation is called a **deadlock**.
 - To handle a deadlock one of T_3 or T_4 must be rolled back and its locks released.

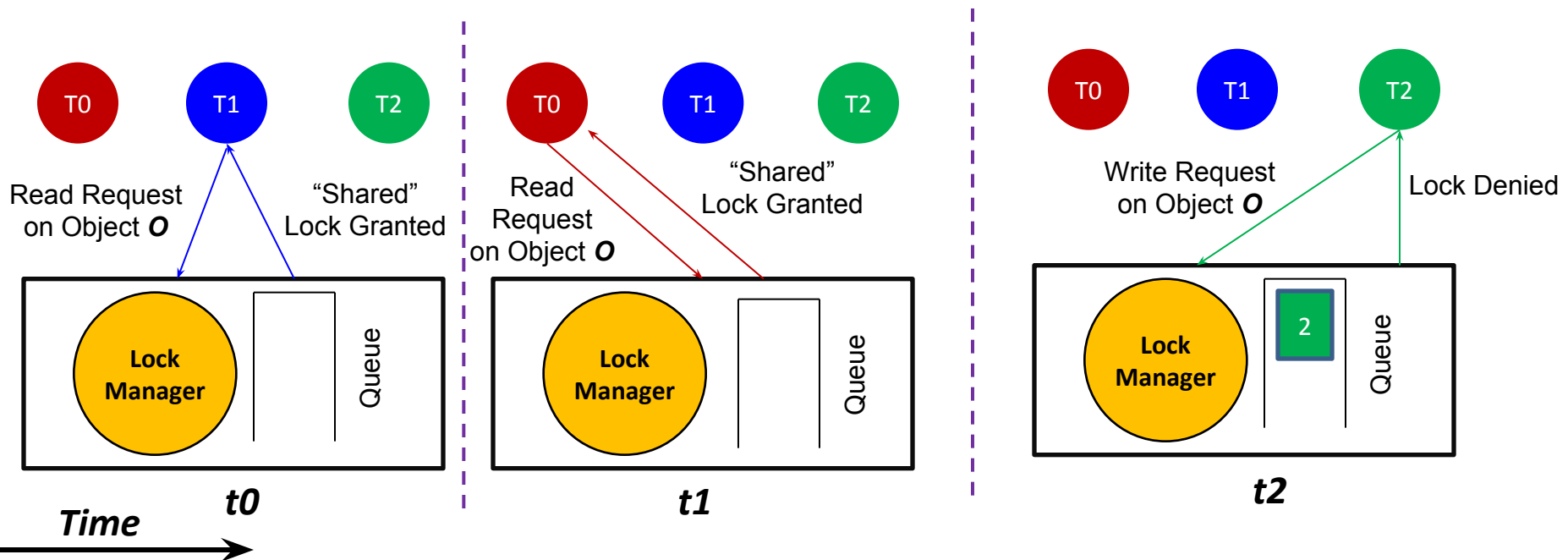


Deadlock (Cont.)

- The potential for deadlock exists in most locking protocols. Deadlocks are a necessary evil.
- **Starvation** is also possible if concurrency control manager is badly designed. For example:
 - A transaction may be waiting for an X-lock on an item, while a sequence of other transactions requests are granted an S-lock on the same item.
 - The same transaction is repeatedly rolled back due to deadlocks.
- Concurrency control manager can be designed to prevent starvation.
- Starvation of transactions can be avoided by granting the locks as follows:
 - When T_i requests a lock on a data item Q in a particular mode M , the concurrency control manager grants the lock provided that
 - *There is no other transaction holding a lock on Q in a mode that conflicts with M*
 - *There is no other transaction that is waiting for a lock on Q and that made its lock request before T_i*

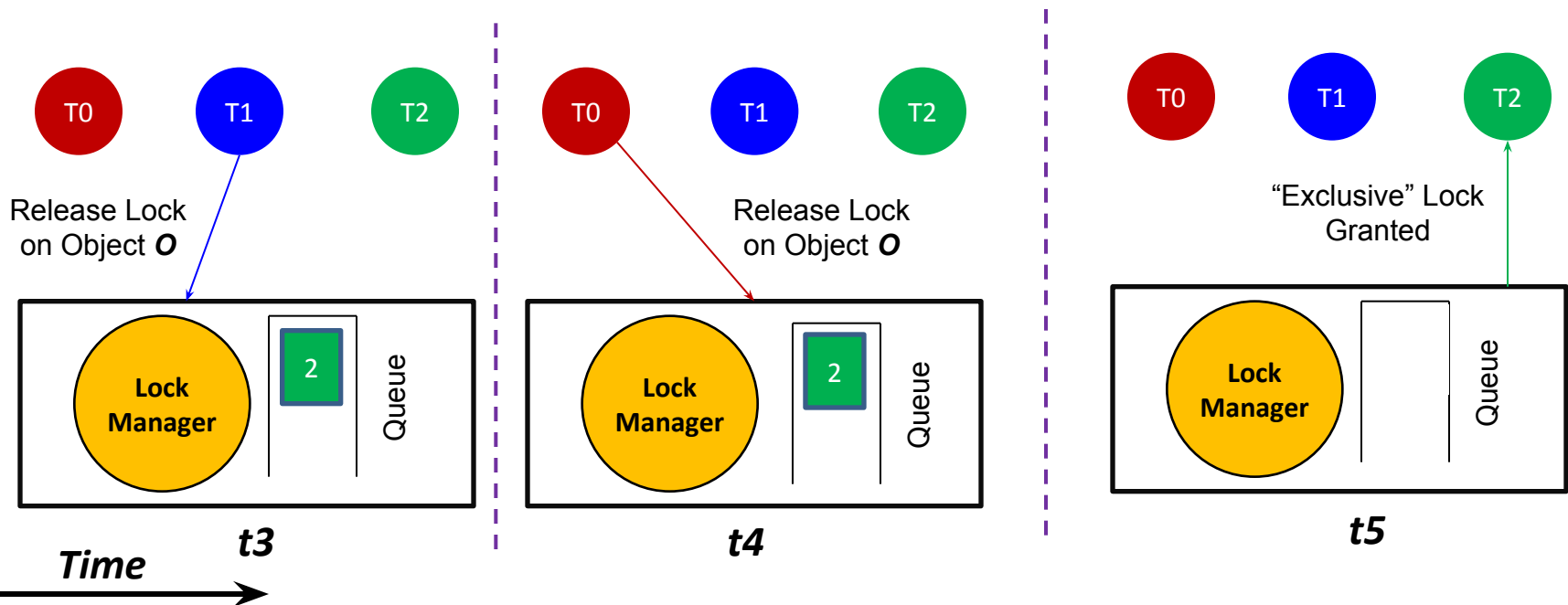
Two-Phase Locking

- A widely used locking protocol, called *Two-Phase Locking* (*2PL*), has two rules:
 - **Rule 1**: if a transaction T wants to read (or write) an object O , it first requests the lock manager for a shared (or exclusive) lock on O



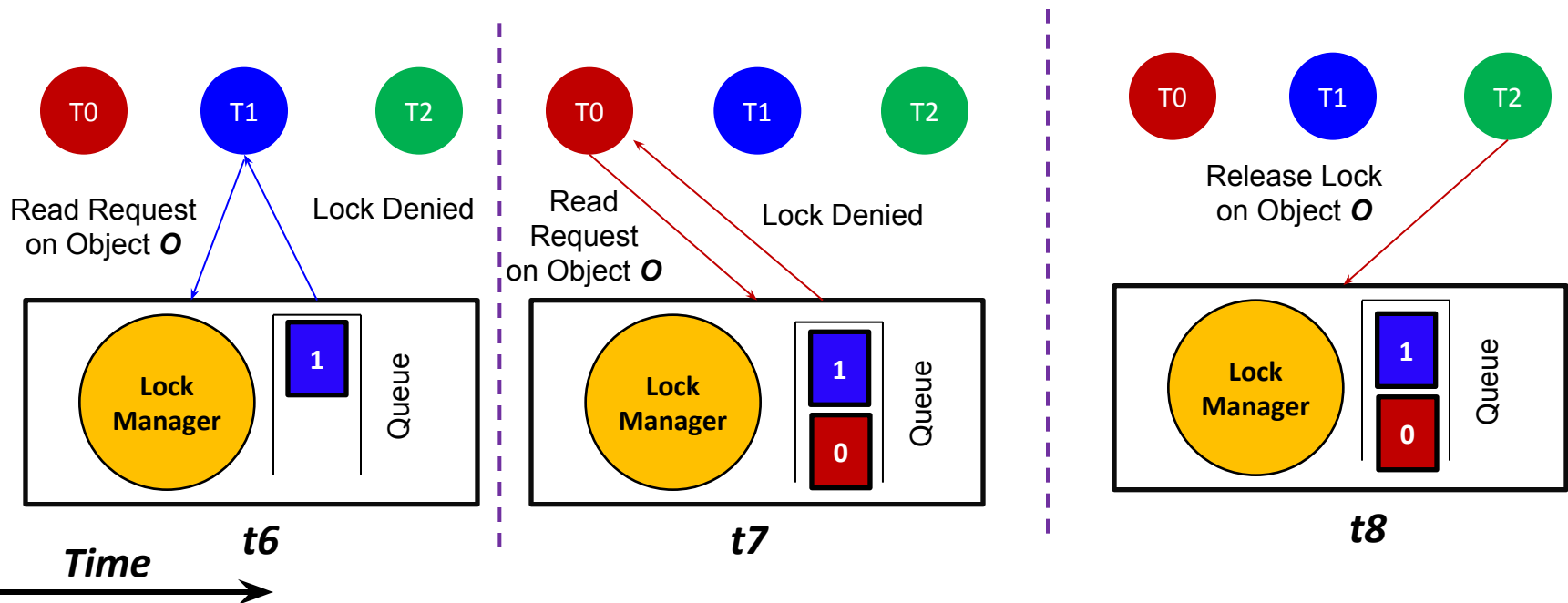
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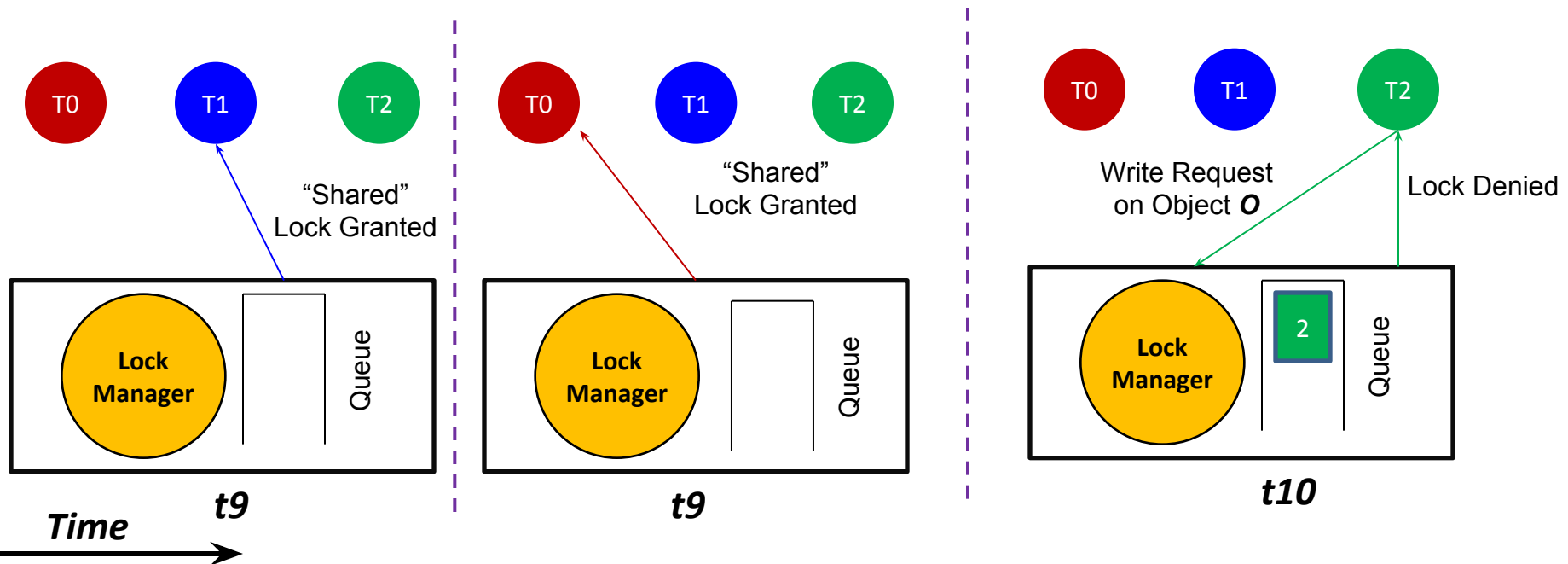
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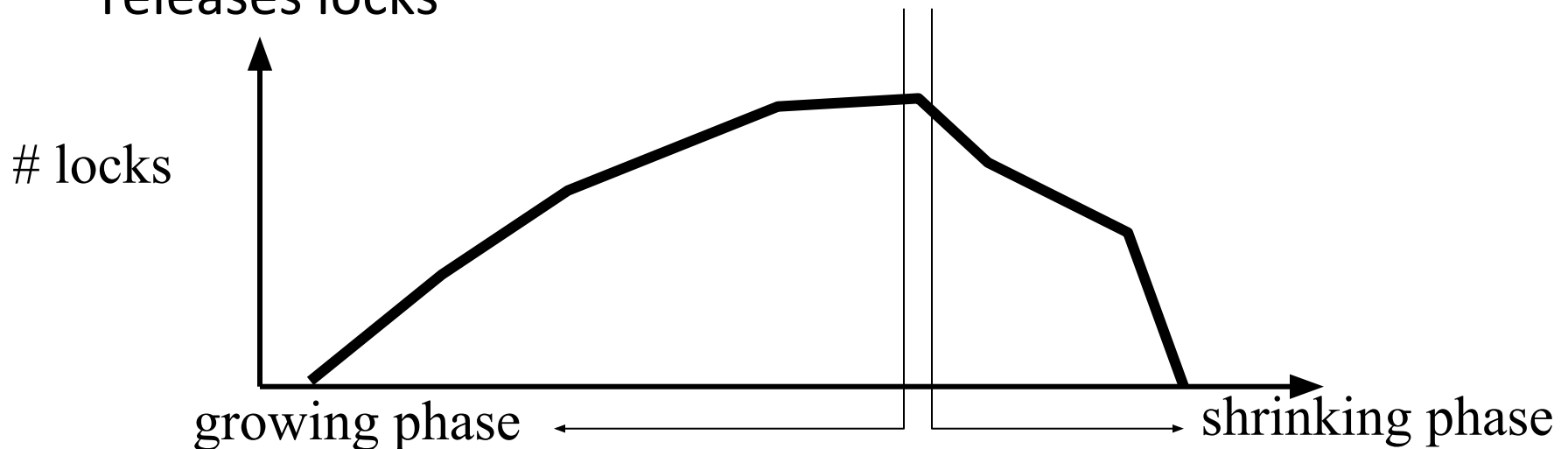
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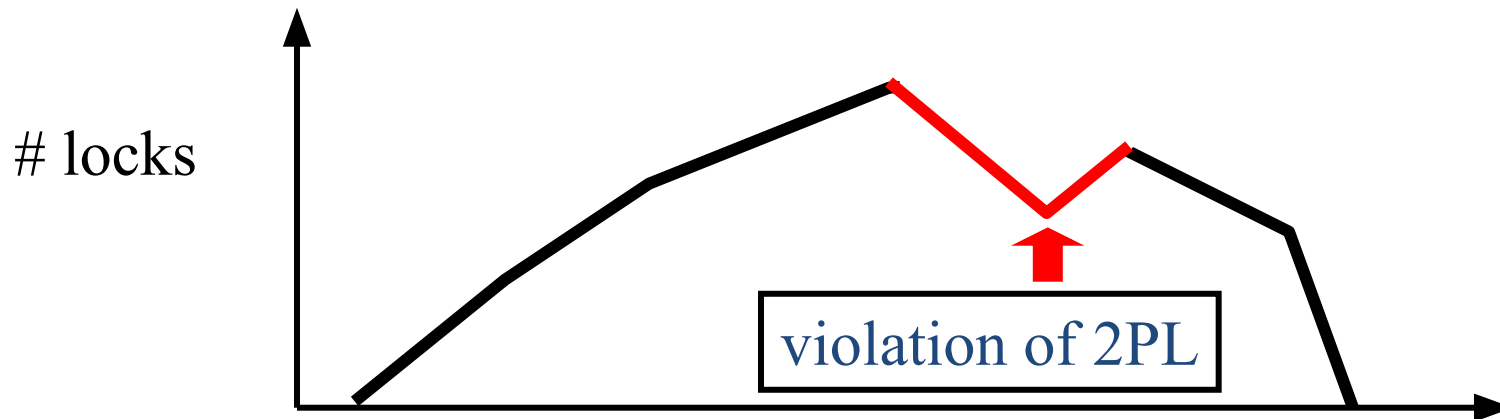
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 - **Rule 2:** *T* can release locks before it *commits* or *aborts*, and cannot request additional locks once it releases any lock
- Thus, every transaction has a “growing” phase in which it acquires locks, followed by a “shrinking” phase in which it releases locks



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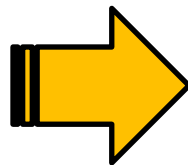


Resolving RW Conflicts Using 2PL

- Suppose that T1 and T2 actions are interleaved as follows:
 - T1 reads A
 - T2 reads A, decrements A and commits
 - T1 tries to decrement A
- T1 and T2 can be represented by the following schedule:

T1	T2
R(A)	R(A)
	W(A)
	Commit
W(A)	
Commit	

Exposes RW Anomaly



T1	T2
EXCLUSIVE(A)	
R(A)	
W(A)	
Commit	
	Lock(A)
	Wait
	EXCLUSIVE(A)
	R(A)
	W(A)
	Commit

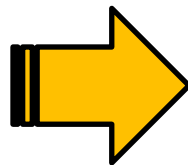
RW
Conflict
Resolved!

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	R(A)
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	Commit
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Commit	

Exposes RW Anomaly



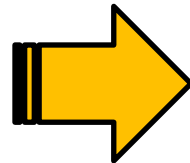
T1	T2
EXCLUSIVE(A)	
R(A)	
W(A)	
Commit	
	Lock(A)
	Wait
	EXCLUSIVE(A)
	R(A)
	W(A)
	Commit

But, it can
limit
parallelism!

Resolving WW Conflicts Using 2PL

- Suppose that T1 and T2 actions are interleaved as follows:
 - T1 sets Ravi's Salary to \$1000
 - T2 sets Kishan's Salary to \$2000
 - T1 sets Kishan's Salary to \$1000
 - T2 sets Ravi's Salary to \$2000
- T1 and T2 can be represented by the following schedule:

T1	T2
W(RS)	
	W(KS)
W(KS)	
Commit	W(RS)
	Commit



T1	T2
EXCLUSIVE(RS)	
EXCLUSIVE(KS)	
W(RS)	
W(KS)	
Commit	
	Lock(KS)
	Wait
	EXCLUSIVE(KS)
	EXCLUSIVE(RS)
	W(KS)
	W(RS)
	Commit

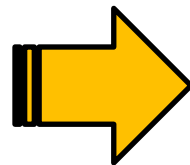
WW
Conflict
Resolved!

Exposes WW Anomaly
(assuming, RS & KS must be kept equal)

Resolving WW Conflicts Using 2PL

- Suppose that T1 and T2 actions are interleaved as follows:
 - T1 sets Ravi's Salary to \$1000
 - T2 sets Kishan's Salary to \$2000
 - T1 sets Kishan's Salary to \$1000
 - T2 sets Ravi's Salary to \$2000
- T1 and T2 can be represented by the following schedule:

T1	T2
W(RS)	W(KS)
W(KS)	W(RS)
Commit	Commit



T1	T2
EXCLUSIVE(RS)	
W(RS)	
Lock(KS)	EXCLUSIVE(KS)
↑	W(KS)
Wait	Lock(RS)
	↑
	Wait

Exposes WW Anomaly

(assuming, MS & AS must be kept equal)

Deadlock!



Cascading rollback may occur under two-phase locking

T5	T6	T7
lock-X(A)		
read(A)		
lock-S(B)		
read(B)		
write(A)		
unlock(A)		
	lock-X(A)	
	read(A)	
	write(A)	
	unlock(A)	
		lock-S(A)
		read(A)
abort		

Cascading rollback may occur under two-phase locking.

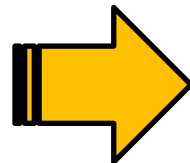
Resolving WR Conflicts

- Suppose that T1 and T2 actions are *interleaved* as follows:
 - T1 deducts \$100 from account A
 - T2 adds 6% interest to accounts A and B
 - T1 credits \$100 to account B
- T1 and T2 can be represented by the following schedule:

T1	T2
R(A)	
W(A)	
	R(A)
	W(A)
	R(B)
	W(B)
	Commit

R(B)
W(B)
Commit

Exposes WR Anomaly

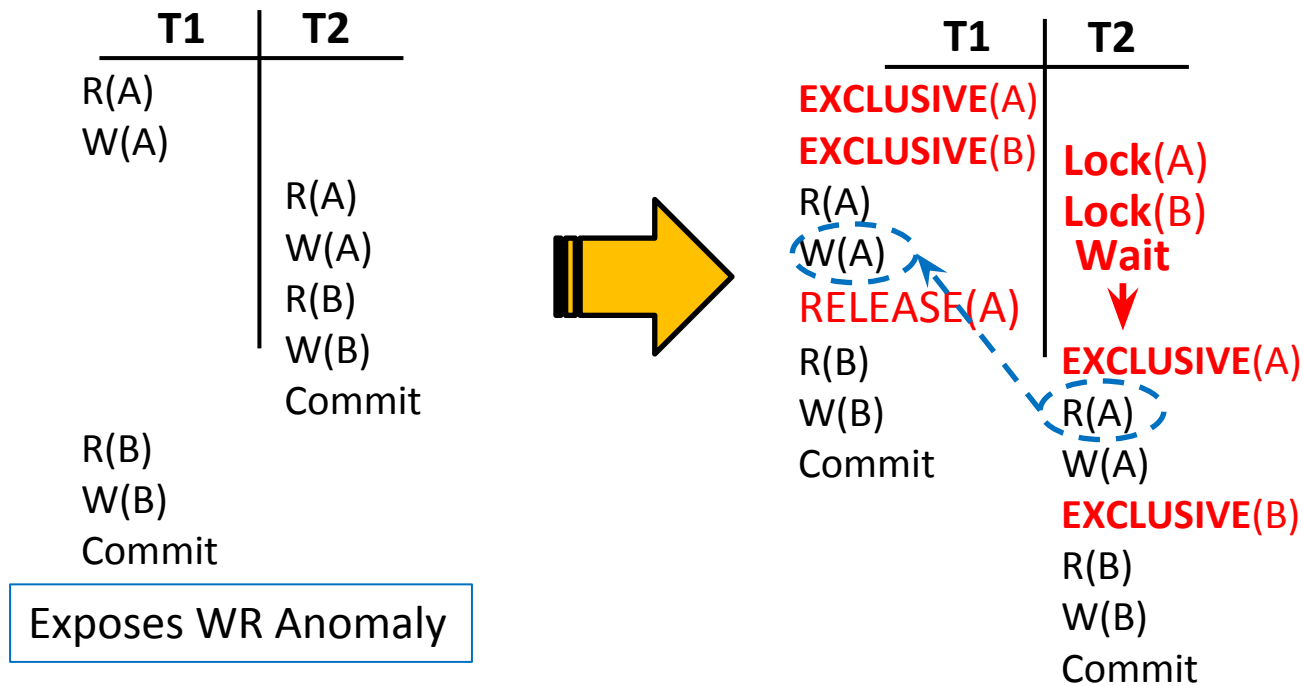


T1	T2
EXCLUSIVE(A)	
EXCLUSIVE(B)	
R(A)	
W(A)	
R(B)	
W(B)	
Commit	
	Lock(A)
	Lock(B)
	Wait
	EXCLUSIVE(A)
	EXCLUSIVE(B)
	R(A)
	W(A)
	R(B)
	W(B)
	Commit

WR
Conflict
Resolved!

Resolving WR Conflicts

- Suppose that T1 and T2 actions are *interleaved* as follows:
 - T1 deducts \$100 from account A
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 - T1 credits \$100 to account B
- T1 and T2 can be represented by the following schedule:



WR
Conflict is
NOT
Resolved!

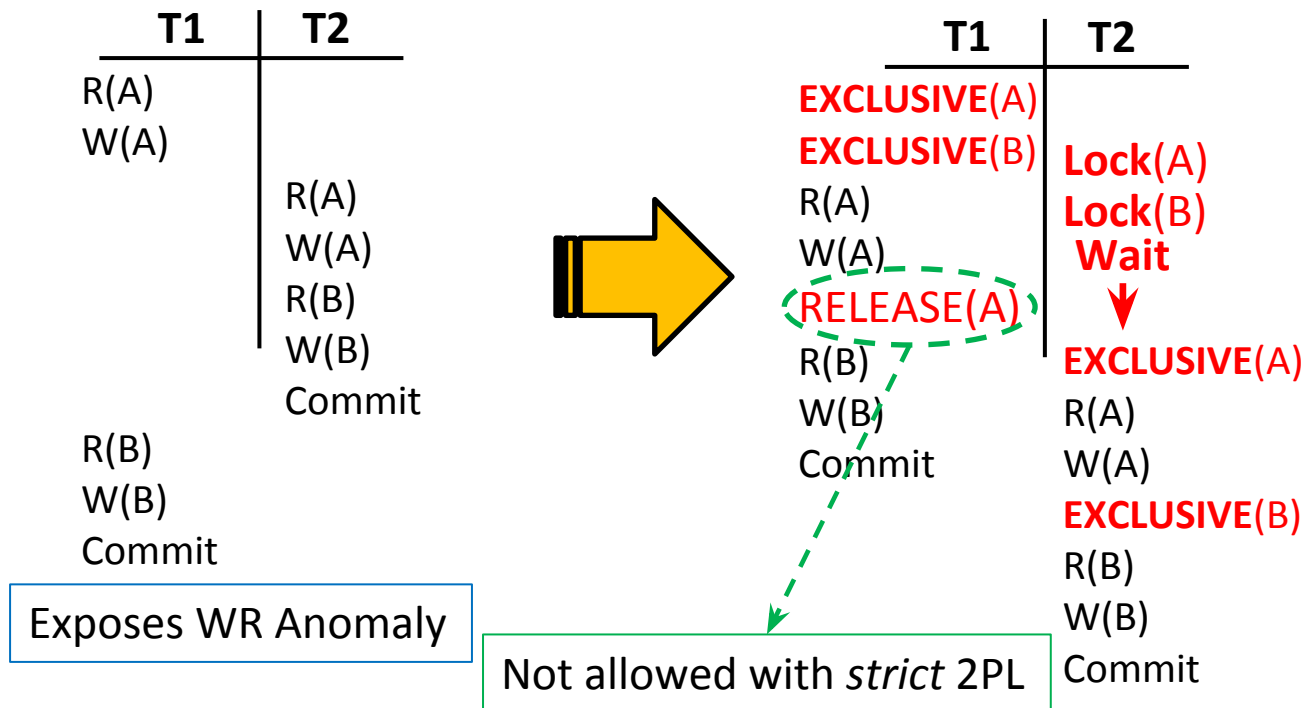
How can
we solve
this?

Strict Two-Phase Locking

- WR conflicts (as well as RW & WW) can be solved by making 2PL *stricter*
- In particular, *Rule 2* in 2PL can be modified as follows:
 - *Rule 2*: locks of a transaction T can only be released after T completes (i.e., commits or aborts)
- This version of 2PL is called *Strict Two-Phase Locking*

Resolving WR Conflicts: *Revisit*

- Suppose that T1 and T2 actions are *interleaved* as follows:
 - T1 deducts \$100 from account A
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 - T1 credits \$100 to account B
- T1 and T2 can be represented by the following schedule:



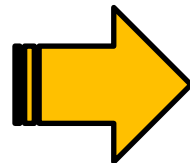
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T1	T2
R(A)	
W(A)	
	R(A)
	W(A)
	R(B)
	W(B)
	Commit

R(B)
W(B)
Commit

Exposes WR Anomaly



T1	T2
EXCLUSIVE(A)	
EXCLUSIVE(B)	
R(A)	
W(A)	
R(B)	
W(B)	
Commit	
	Lock(A)
	Lock(B)
	Wait
	EXCLUSIVE(A)
	EXCLUSIVE(B)
	R(A)
	W(A)
	R(B)
	W(B)
	Commit

WR Conflict
is Resolved!

But,
parallelism
is limited
more!

2PL vs. Strict 2PL

- Two-Phase Locking (2PL):

- Limits concurrency
- May lead to deadlocks
- May have 'dirty reads'

- Strict 2PL:

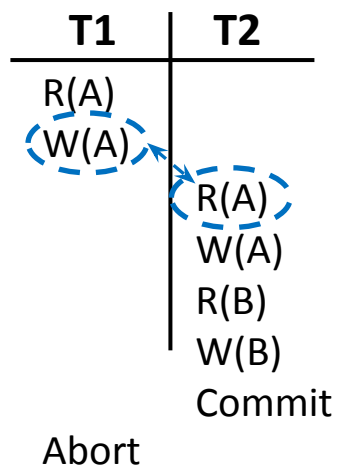
- Limits concurrency more (*but*, actions of different transactions can still be interleaved)
- May still lead to deadlocks
- Avoids 'dirty reads'

T1	T2
SHARED(A) R(A)	
	SHARED(A) R(A)
	EXCLUSIVE(B) R(B)
EXCLUSIVE(C) R(C) W(C) Commit	W(B) Commit

A Schedule with *Strict 2PL* and *Interleaved Actions*

Schedules with *Aborted* Transactions

- Suppose that T1 and T2 actions are interleaved as follows:
 - T1 deducts \$100 from account A
 - T2 adds 6% interest to accounts A and B, and commits
 - T1 is aborted
- T1 and T2 can be represented by the following schedule:

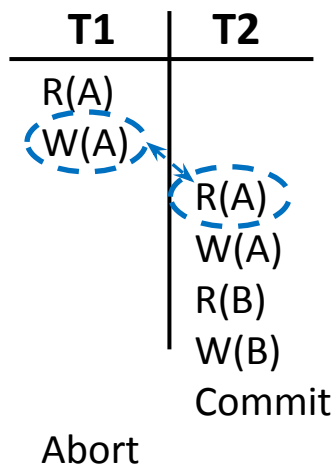


T2 read a value for A that should have never been there!

How can we deal with the situation, assuming T2 had not yet committed?

Schedules with *Aborted* Transactions

- Suppose that T1 and T2 actions are interleaved as follows:
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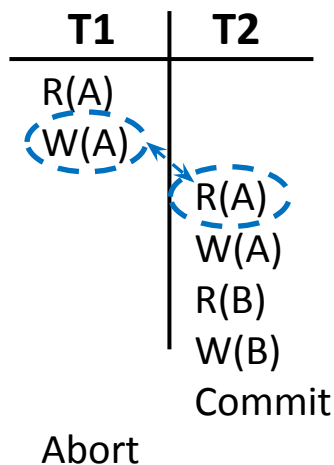
T2 read a value for A that should have never been there!

We can *cascade* the abort of T1 by aborting T2 as well!

This “cascading process” can be *recursively* applied to any transaction that read A written by T1

Schedules with *Aborted* Transactions

- Suppose that T1 and T2 actions are interleaved as follows:
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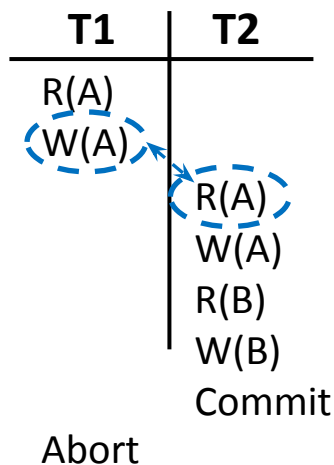
T2 read a value for A that should have never been there!

How can we deal with the situation, assuming T2 had actually committed?

The schedule is indeed unrecoverable!

Schedules with *Aborted* Transactions

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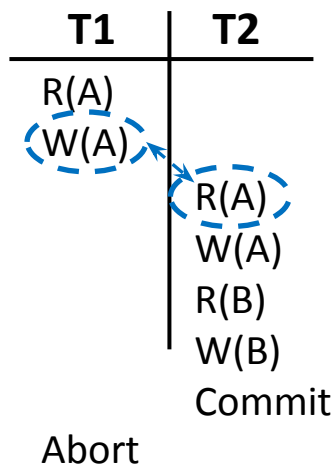
T2 read a value for A that should have never been there!

For a schedule to be *recoverable*, transactions should commit only after all transactions whose changes they read commit!

“Recoverable schedules” avoid *cascading aborts*!

Schedules with *Aborted* Transactions

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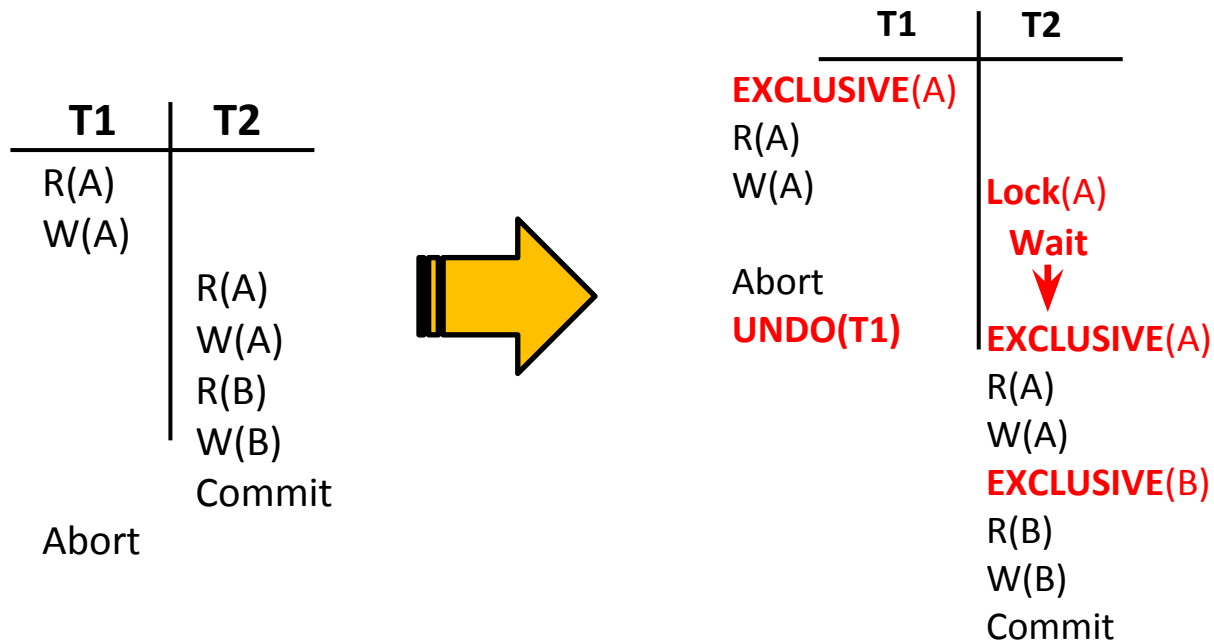
T2 read a value for A that should have never been there!

How can we ensure “recoverable schedules”?

By using Strict 2PL!

Schedules with *Aborted* Transactions

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Cascaded
aborts are
avoided!

Serializable Schedules: *Redefined*

- Two schedules are said to be *equivalent* if for any database state, the effect of executing the 1st schedule is identical to the effect of executing the 2nd schedule
- Previously: a *serializable schedule* is a schedule that is equivalent to a serial schedule
- Now: a *serializable schedule* is a schedule that is equivalent to a serial schedule over a set of committed transactions
- This definition captures *serializability* as well as *recoverability*

Lock Conversions

- A transaction may need to change the lock it already acquires on an object
 - From Shared to Exclusive
 - This is referred to as *lock upgrade*
 - From Exclusive to Shared
 - This is referred to as *lock downgrade*
- This protocol ensures serializability
- For example, an SQL update statement might acquire a Shared lock on each row, ***R***, in a table and if ***R*** satisfies the condition (in the WHERE clause), an Exclusive lock must be obtained for ***R***

Lock Upgrades

- A lock upgrade request from a transaction T on object O must be handled specially by:
 - Granting an Exclusive lock to T immediately *if no other transaction holds a lock on O*
 - Otherwise, queuing T at the front of O 's queue (i.e., T is favored)
- T is *favored* because it already holds a Shared lock on O
 - Queuing T *in front of* another transaction T' that holds no lock on O , but requested an Exclusive lock on O averts a deadlock!
 - However, if T and T' hold a Shared lock on O , and both request upgrades to an Exclusive lock, a deadlock will arise regardless!

Lock Downgrades

- Lock upgrades can be entirely avoided by obtaining Exclusive locks *initially*, and downgrade them to Shared locks once needed
- Would this violate any 2PL requirement?
 - On the surface yes; since the transaction (say, T) may need to upgrade later
 - This is a special case as T conservatively obtained an Exclusive lock, and did nothing but read the object that it downgraded
 - 2PL can be safely extended to allow lock downgrades in the growing phase, provided that the transaction has not modified the object



Automatic Acquisition of Locks

- A transaction T_i issues the standard read/write instruction, without explicit locking calls.
- The operation **read**(Q) is processed as:

```
    if no lock on  $Q$ 
    then
        grant  $T_i$  a lock-S on  $Q$ ;
read( $Q$ )
    else begin
        if necessary wait until no other
            transaction has a lock-X on  $Q$ 
        grant  $T_i$  a lock-S on  $Q$ ;
        read( $Q$ )
    end
```



Automatic Acquisition of Locks (Cont.)

- The operation **write(Q)** is processed as:
if *no* lock on Q
 then
 grant T_i a **lock-X** on Q
 write(Q)
 else begin
 if necessary wait until no other trans. has any lock on Q ,
 if T_i has a **lock-S** on Q
 then
 upgrade lock on Q to **lock-X**
 else
 grant T_i a **lock-X** on Q
 write(Q)
 end;
- All locks are released after commit or abort



Implementation of Locking

- A **lock manager** can be implemented as a separate process
- Transactions can send lock and unlock requests as messages
- The lock manager replies to a lock request by sending a lock grant messages (or a message asking the transaction to roll back, in case of a deadlock)
 - The requesting transaction waits until its request is answered
- The lock manager maintains an in-memory data-structure called a **lock table** to record granted locks and pending requests

Lock Manager Implementation

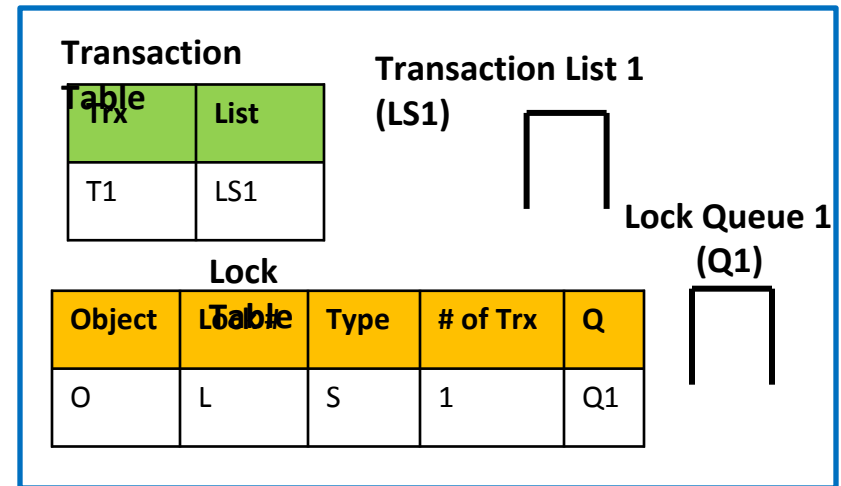
- Usually, a lock manager in a DBMS maintains three types of data structures:

- A queue, **Q**, for each lock, **L**, to hold its pending requests

- A lock table, which keeps for each **L** associated with each object, **O**, a record **R** that contains:

- The type of **L** (e.g., shared or exclusive)
 - The number of transactions currently holding **L** on **O**
 - A pointer to **Q**

- A transaction table, which maintains for each transaction, **T**, a pointer to a list of locks held by **T**





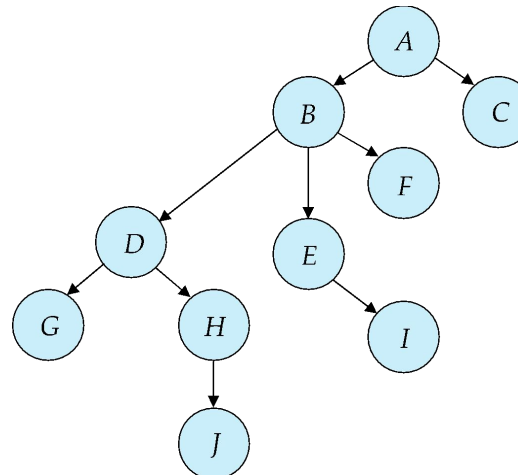
Graph-Based Protocols

- Graph-based protocols are an alternative to two-phase locking
- Impose a partial ordering $\rightarrow$ on the set $\mathbf{D} = \{d_1, d_2, \dots, d_h\}$ of all data items.
 - If $d_i \rightarrow d_j$ then any transaction accessing both d_i and d_j must access d_i before accessing d_j .
 - Implies that the set $\mathbf{D}$ may now be viewed as a directed acyclic graph, called a *database graph*.
- The *tree-protocol* is a simple kind of graph protocol.



Tree Protocol

- Only exclusive locks are allowed.
- The first lock by T_i may be on any data item. Subsequently, a data Q can be locked by T_i only if the parent of Q is currently locked by T_i .
- Data items may be unlocked at any time.
- A data item that has been locked and unlocked by T_i cannot subsequently be relocked by T_i .





Graph-Based Protocols (Cont.)

- The tree protocol ensures conflict serializability as well as freedom from deadlock.
- Unlocking may occur earlier in the tree-locking protocol than in the two-phase locking protocol.
 - Shorter waiting times, and increase in concurrency
 - Protocol is deadlock-free, no rollbacks are required
- Drawbacks
 - Protocol does not guarantee recoverability or cascade freedom
 - Need to introduce commit dependencies to ensure recoverability
 - Transactions may have to lock data items that they do not access.
 - increased locking overhead, and additional waiting time
 - potential decrease in concurrency
- Schedules not possible under two-phase locking are possible under the tree protocol, and vice versa.



Deadlock Handling

- System is **deadlocked** if there is a set of transactions such that every transaction in the set is waiting for another transaction in the set.

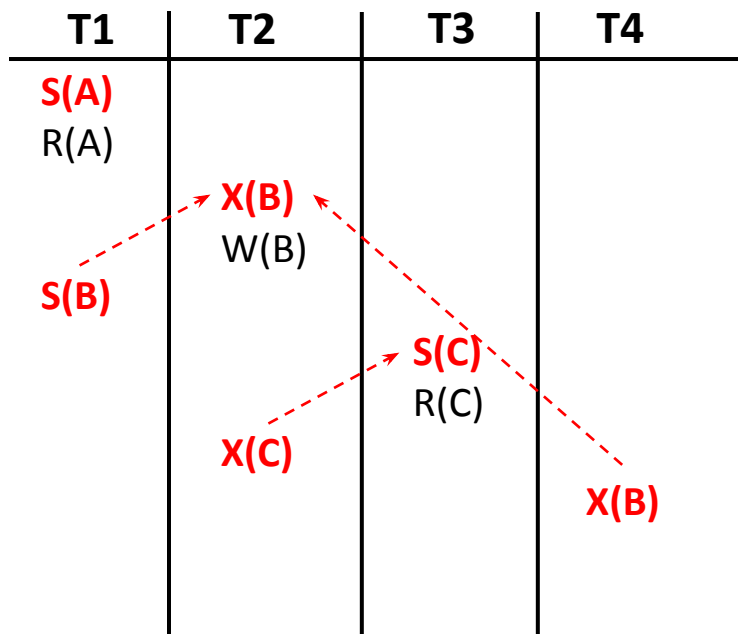
T_3	T_4
lock-X(B) read(B) $B := B - 50$ write(B)	
	lock-S(A) read(A) lock-S(B)
lock-X(A)	

Deadlock Detection

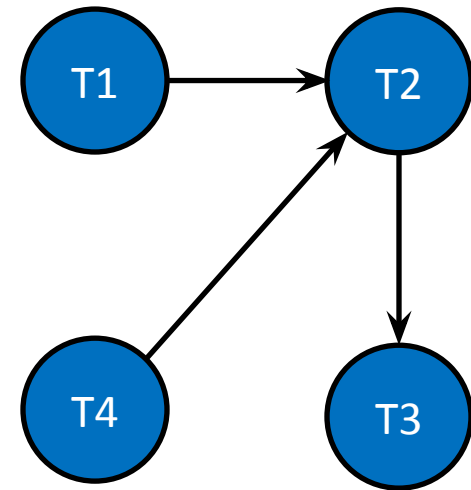
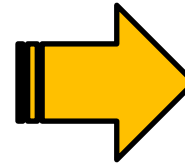
- The lock manager maintains a structure called a *waits-for graph* to *periodically* detect deadlocks
- In a waits-for graph:
 - The nodes correspond to active transactions
 - There is an edge from T_i to T_j *if and only if* T_i is waiting for T_j to release a lock
- The lock manager *adds* and *removes* edges to and from a waits-for graph when it *queues* and *grants* lock requests, respectively
- A deadlock is detected when a *cycle* in the waits-for graph is found

Deadlock Detection (*Cont'd*)

- The following schedule is free of deadlocks:



A schedule without a deadlock



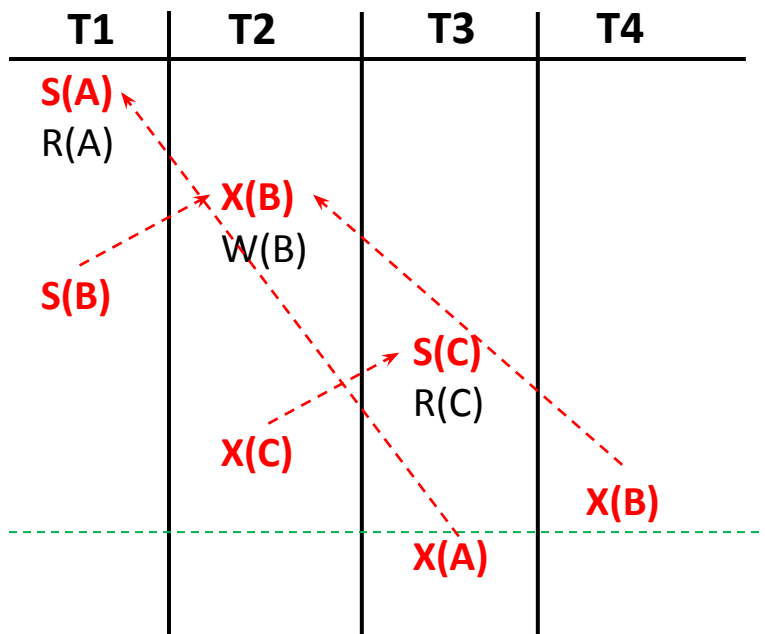
No cycles; hence, no deadlocks!

The Corresponding **Waits-For Graph***

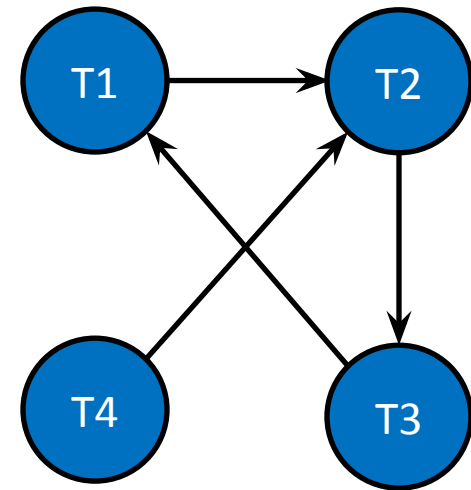
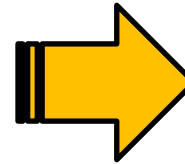
*The nodes correspond to active transactions and there is an edge from T_i to T_j *if and only if* T_i is waiting for T_j to release a lock

Deadlock Detection (*Cont'd*)

- The following schedule is **NOT** free of deadlocks:



A schedule with a deadlock

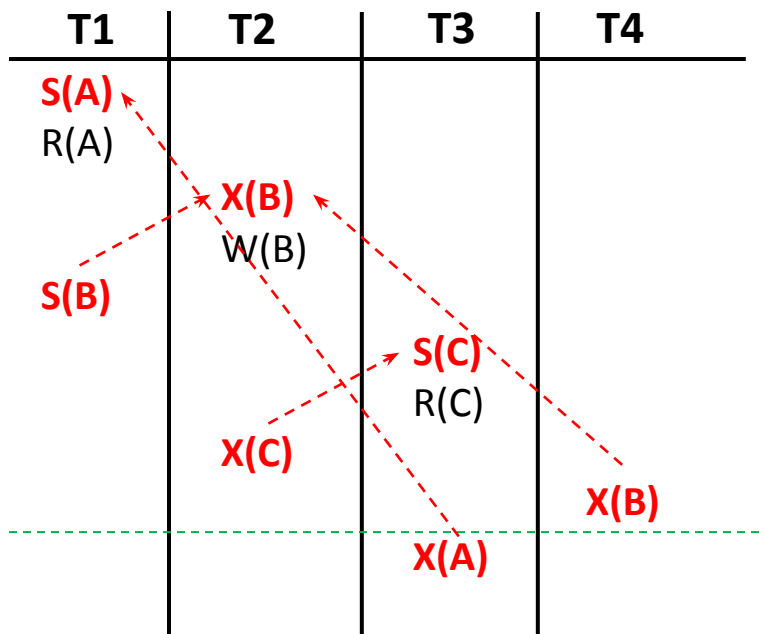


The Corresponding **Waits-For Graph***

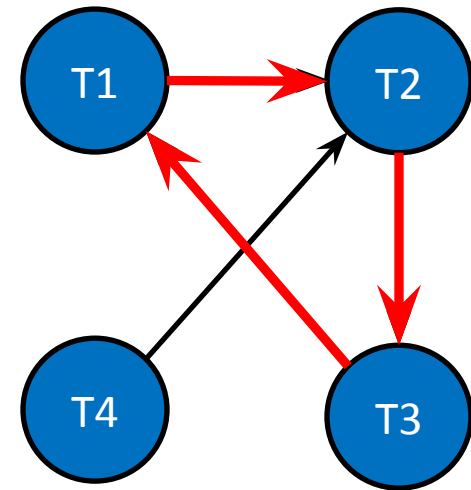
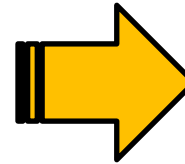
*The nodes correspond to active transactions and there is an edge from T_i to T_j *if and only if* T_i is waiting for T_j to release a lock

Deadlock Detection (*Cont'd*)

- The following schedule is **NOT** free of deadlocks:



A schedule with a deadlock



Cycle detected; hence, a deadlock!

The Corresponding **Waits-For Graph***

*The nodes correspond to active transactions and there is an edge from T_i to T_j *if and only if* T_i is waiting for T_j to release a lock

Resolving Deadlocks

- A deadlock is resolved by aborting a transaction that is on a cycle and releasing its locks
 - This allows some of the waiting transactions to proceed
- When deadlock is detected :
 - Some transaction will have to rolled back (made a **victim**) to break deadlock cycle.
 - Select that transaction as victim that will incur minimum cost
 - Rollback -- determine how far to roll back transaction
 - **Total rollback**: Abort the transaction and then restart it.
 - **Partial rollback**: Roll back victim transaction only as far as necessary to release locks that another transaction in cycle is waiting for
- The choice of which transaction to abort can be made using different criteria:
 - The one with the fewest locks
 - Or the one that has done the least work
 - Or the one that is farthest from completion (*more accurate*)
- Starvation can happen (why?)
 - One solution: oldest transaction in the deadlock set is never chosen as victim
 - **Caveat**: a transaction that was aborted in the past, should be *favored* subsequently and not aborted upon a deadlock detection!

Deadlock Prevention

- Studies indicate that deadlocks are relatively infrequent and *detection-based schemes* work well in practice
- However, if there is a high level of *contention* for locks, *prevention-based schemes* could perform better
- Deadlocks can be averted by giving each transaction a *priority* and ensuring that lower-priority transactions are not allowed to wait for higher-priority ones (or vice versa)



Deadlock Prevention Strategies

- One way to assign priorities is to give each transaction a *timestamp* when it is started
 - Thus, the lower the timestamp, the higher is the transaction's priority, i.e. older the transaction.
- If a transaction T_i requests a lock and a transaction T_j holds a conflicting lock, the lock manager can use one of the following policies:
 - **Wait-Die scheme** (non-preemptive): **If T_i is older than T_j , T_i is allowed to wait; otherwise, T_j is rolled back (dies).** For example, T_1 , T_2 , and T_3 have timestamps 5, 10, and 15, respectively. If T_1 requests a data item held by T_2 , then T_1 will wait, If T_3 requests a data item held by T_2 , then T_3 will be rolled back.
 - Older transaction may wait for younger one to release data item.
 - Younger transactions never wait for older ones; they are rolled back instead.
 - A transaction may die several times before acquiring a lock
 - **Wound-Wait scheme**(preemptive): **If T_i is older than T_j , T_j is rolled back (i.e. T_j is wounded by T_i); otherwise, T_j is allowed to wait.** For example, T_1 , T_2 , and T_3 have timestamps 5, 10, and 15, respectively. If T_1 requests a data item held by T_2 , then T_2 will be rolled back. If T_3 requests a data item held by T_2 , then T_3 will wait.
 - Older transaction *wounds* of younger transaction instead of waiting for it.
 - Younger transactions may wait for older ones.
 - Fewer rollbacks than *wait-die* scheme.
- In both schemes, a rolled back transactions is restarted with its original timestamp.
 - Ensures that older transactions have precedence over newer ones, and starvation is thus avoided.



Deadlock prevention (Cont.)

■ Timeout-Based Schemes:

- A transaction waits for a lock only for a specified amount of time. After that, the wait times out and the transaction is rolled back.
- Ensures that deadlocks get resolved by timeout if they occur
- Simple to implement
- But may roll back transaction unnecessarily in absence of deadlock
 - Difficult to determine good value of the timeout interval.
- Starvation is also possible



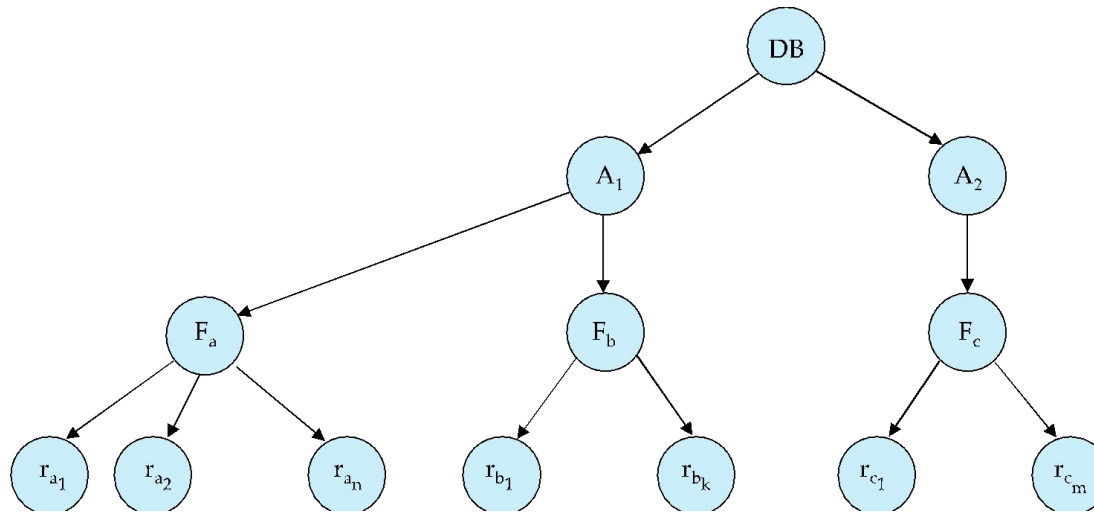
Multiple Granularity

- Allow data items to be of various sizes and define a hierarchy of data granularities, where the small granularities are nested within larger ones
- Can be represented graphically as a tree
- When a transaction locks a node in the tree *explicitly*, it *implicitly* locks all the node's descendants in the same mode.
- Granularity of locking (level in tree where locking is done):
 - **Fine granularity** (lower in tree): high concurrency, high locking overhead
 - **Coarse granularity** (higher in tree): low locking overhead, low concurrency



Example of Granularity Hierarchy

- The levels, starting from the coarsest (top) level are
 - *database*
 - *area*
 - *file*
 - *record*
- The corresponding tree





Intention Lock Modes

- In addition to S and X lock modes, there are three additional lock modes with multiple granularity:
 - ***intention-shared*** (IS): indicates explicit locking at a lower level of the tree but only with shared locks.
 - ***intention-exclusive*** (IX): indicates explicit locking at a lower level with exclusive or shared locks
 - ***shared and intention-exclusive*** (SIX): the subtree rooted by that node is locked explicitly in shared mode and explicit locking is being done at a lower level with exclusive-mode locks.
- Intention locks allow a higher level node to be locked in S or X mode without having to check all descendent nodes.



Compatibility Matrix with Intention Lock Modes

- The compatibility matrix for all lock modes is:

	IS	IX	S	SIX	X
IS	true	true	true	true	false
IX	true	true	false	false	false
S	true	false	true	false	false
SIX	true	false	false	false	false
X	false	false	false	false	false



Multiple Granularity Locking Scheme

- Transaction T_i can lock a node Q , using the following rules:
 1. The lock compatibility matrix must be observed.
 2. The root of the tree must be locked first, and may be locked in any mode.
 3. A node Q can be locked by T_i in S or IS mode only if the parent of Q is currently locked by T_i in either IX or IS mode.
 4. A node Q can be locked by T_i in X, SIX, or IX mode only if the parent of Q is currently locked by T_i in either IX or SIX mode.
 5. T_i can lock a node only if it has not previously unlocked any node (that is, T_i is two-phase).
 6. T_i can unlock a node Q only if none of the children of Q are currently locked by T_i .
- Observe that locks are acquired in root-to-leaf order, whereas they are released in leaf-to-root order.
- **Lock granularity escalation**: in case there are too many locks at a particular level, switch to higher granularity S or X lock



Insert/Delete Operations and Predicate Reads

- Locking rules for insert/delete operations
 - An exclusive lock must be obtained on an item before it is deleted
 - A transaction that inserts a new tuple into the database is automatically given an X-mode lock on the tuple
- Ensures that
 - reads/writes conflict with deletes
 - Inserted tuple is not accessible by other transactions until the transaction that inserts the tuple commits



Phantom Phenomenon

- Example of **phantom phenomenon**.
 - A transaction T1 that performs **predicate read** (or scan) of a relation
 - **select count(*)**
from *instructor*
where *dept_name* = 'Physics'
 - and a transaction T2 that inserts a tuple while T1 is active but after predicate read
 - **insert into** *instructor* **values** ('11111', 'Feynman', 'Physics', 94000)
- If only tuple locks are used, non-serializable schedules can result
 - E.g. the scan transaction does not see the new instructor, but may read some other tuple written by the update transaction



Insert/Delete Operations and Predicate Reads

- **Another Example:** T1 and T2 both find maximum instructor ID in parallel, and create new instructors with $ID = \text{maximum ID} + 1$
 - Both instructors get same ID, not possible in serializable schedule
- Schedule

T1	T2
Read(instructor where dept_name='Physics')	
	Insert Instructor in Physics
	Insert Instructor in Comp. Sci.
	Commit
Read(instructor where dept_name='Comp. Sci.')	



Handling Phantoms

- There is a conflict at the data level
 - The transaction performing predicate read or scanning the relation is reading information that indicates what tuples the relation contains
 - The transaction inserting/deleting/updating a tuple updates the same information.
 - The conflict should be detected, e.g. by locking the information.
- One solution:
 - Associate a data item with the relation, to represent the information about what tuples the relation contains.
 - Transactions scanning the relation acquire a shared lock in the data item,
 - Transactions inserting or deleting a tuple acquire an exclusive lock on the data item. (Note: locks on the data item do not conflict with locks on individual tuples.)
- Above protocol provides very low concurrency for insertions/deletions.



Index Locking To Prevent Phantoms

- **Index locking protocol** to prevent phantoms
 - Every relation must have at least one index.
 - A transaction can access tuples only after finding them through one or more indices on the relation
 - A transaction T_i that performs a lookup must lock all the index leaf nodes that it accesses, in S-mode
 - Even if the leaf node does not contain any tuple satisfying the index lookup (e.g. for a range query, no tuple in a leaf is in the range)
 - A transaction T_i that inserts, updates or deletes a tuple t_i in a relation r
 - Must update all indices to r
 - Must obtain exclusive locks on all index leaf nodes affected by the insert/update/delete
 - The rules of the two-phase locking protocol must be observed
- Guarantees that phantom phenomenon won't occur



Next-Key Locking to Prevent Phantoms

- Index-locking protocol to prevent phantoms locks entire leaf node
 - Can result in poor concurrency if there are many inserts
- **Next-key locking protocol**: provides higher concurrency
 - Lock all values that satisfy index lookup (match lookup value, or fall in lookup range)
 - Also lock next key value in index
 - even for inserts/deletes
 - Lock mode: S for lookups, X for insert/delete/update
- Ensures detection of query conflicts with inserts, deletes and updates

Consider B+-tree leaf nodes as below, with query predicate $7 \leq X \leq 16$.
Check what happens with next-key locking when inserting: (i) 15 and (ii) 7





Timestamp Based Concurrency Control



Timestamp-Based Protocols

- Each transaction T_i is issued a timestamp $TS(T_i)$ when it enters the system.
 - Each transaction has a *unique* timestamp
 - Newer transactions have timestamps strictly greater than earlier ones
 - Timestamp could be based on a logical counter
 - Real time may not be unique
 - Can use (wall-clock time, logical counter) to ensure
- Timestamp-based protocols manage concurrent execution such that
time-stamp order = serializability order
- Several alternative protocols based on timestamps



Timestamp-Ordering Protocol

The **timestamp ordering (TSO) protocol**

- Maintains for each data Q two timestamp values:
 - **W-timestamp**(Q) is the largest time-stamp of any transaction that executed **write**(Q) successfully.
 - **R-timestamp**(Q) is the largest time-stamp of any transaction that executed **read**(Q) successfully.
- Imposes rules on read and write operations to ensure that
 - Any conflicting operations are executed in timestamp order
 - Out of order operations cause transaction rollback



Timestamp-Based Protocols (Cont.)

- Suppose a transaction T_i issues a **read**(Q)
 1. If $TS(T_i) \leq \mathbf{W}\text{-timestamp}(Q)$, then T_i needs to read a value of Q that was already overwritten.
 - Hence, the **read** operation is rejected, and T_i is rolled back.
 2. If $TS(T_i) \geq \mathbf{W}\text{-timestamp}(Q)$, then the **read** operation is executed, and $\mathbf{R}\text{-timestamp}(Q)$ is set to
$$\mathbf{max}(\mathbf{R}\text{-timestamp}(Q), TS(T_i)).$$



Timestamp-Based Protocols (Cont.)

- Suppose that transaction T_i issues **write**(Q).
 1. If $TS(T_i) < R\text{-timestamp}(Q)$, then the value of Q that T_i is producing was needed previously, and the system assumed that that value would never be produced.
 - Hence, the **write** operation is rejected, and T_i is rolled back.
 2. If $TS(T_i) < W\text{-timestamp}(Q)$, then T_i is attempting to write an obsolete value of Q .
 - Hence, this **write** operation is rejected, and T_i is rolled back.
 3. Otherwise, the **write** operation is executed, and $W\text{-timestamp}(Q)$ is set to $TS(T_i)$.



Example of Schedule Under TSO

- Is this schedule valid under TSO?

Assume that initially:

$$R-TS(A) = W-TS(A) = 0$$

$$R-TS(B) = W-TS(B) = 0$$

Assume $TS(T_{25}) = 25$ and

$$TS(T_{26}) = 26$$

T_{25}	T_{26}
read(B)	read(B) $B := B - 50$ write(B)
read(A)	read(A)
display($A + B$)	$A := A + 50$ write(A) display($A + B$)

- How about this one,
where initially
 $R-TS(Q) = W-TS(Q) = 0$

T_{27}	T_{28}
read(Q)	write(Q)
write(Q)	



Another Example Under TSO

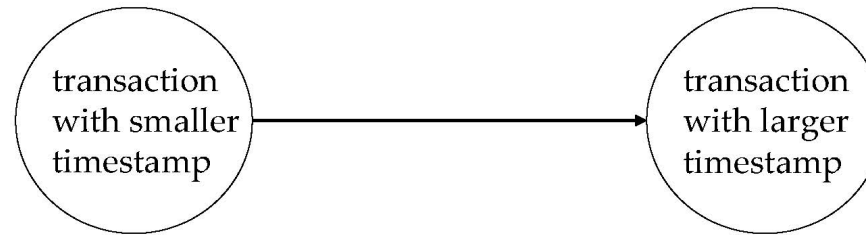
A partial schedule for several data items for transactions with timestamps 1, 2, 3, 4, 5, with all R-TS and W-TS = 0 initially

T_1	T_2	T_3	T_4	T_5
				read (X)
read (Y)	read (Y)	write (Y) write (Z)		
				read (Z)
read (X)	read (Z) abort			
		write (W) abort	read (W)	
				write (Y) write (Z)



Correctness of Timestamp-Ordering Protocol

- The timestamp-ordering protocol guarantees serializability since all the arcs in the precedence graph are of the form:



Thus, there will be no cycles in the precedence graph

- Timestamp protocol ensures freedom from deadlock as no transaction ever waits.
- But the schedule may not be cascade-free, and may not even be recoverable.



Recoverability and Cascade Freedom

- Solution 1:
 - A transaction is structured such that its writes are all performed at the end of its processing
 - All writes of a transaction form an atomic action; no transaction may execute while a transaction is being written
 - A transaction that aborts is restarted with a new timestamp
- Solution 2:
 - Limited form of locking: wait for data to be committed before reading it
- Solution 3:
 - Use commit dependencies to ensure recoverability



Thomas' Write Rule

- Modified version of the timestamp-ordering protocol in which obsolete **write** operations may be ignored under certain circumstances.
- When T_i attempts to write data item Q , if $TS(T_i) < W\text{-timestamp}(Q)$, then T_i is attempting to write an obsolete value of $\{Q\}$.
 - Rather than rolling back T_i as the timestamp ordering protocol would have done, this **{write}** operation can be ignored.
- Otherwise this protocol is the same as the timestamp ordering protocol.
- Thomas' Write Rule allows greater potential concurrency.
 - Allows some view-serializable schedules that are not conflict-serializable.



Validation-Based Protocol

- Idea: can we use commit time as serialization order?
- To do so:
 - Postpone writes to end of transaction
 - Keep track of data items read/written by transaction
 - **Validation** performed at commit time, detect any out-of-serialization order reads/writes
- Also called as **optimistic concurrency control** since transaction executes fully in the hope that all will go well during validation



Validation-Based Protocol

- Execution of transaction T_i is done in three phases.
 1. **Read and execution phase:** Transaction T_i writes only to temporary local variables
 2. **Validation phase:** Transaction T_i performs a "validation test" to determine if local variables can be written without violating serializability.
 3. **Write phase:** If T_i is validated, the updates are applied to the database; otherwise, T_i is rolled back.
- The three phases of concurrently executing transactions can be interleaved, but each transaction must go through the three phases in that order.
 - We assume for simplicity that the validation and write phase occur together, atomically and serially
 - I.e., only one transaction executes validation/write at a time.



Validation-Based Protocol (Cont.)

- Each transaction T_i has 3 timestamps
 - **StartTS**(T_i) : the time when T_i started its execution
 - **ValidationTS**(T_i): the time when T_i entered its validation phase
 - **FinishTS**(T_i) : the time when T_i finished its write phase
- Validation tests use above timestamps and read/write sets to ensure that serializability order is determined by validation time
 - Thus, $TS(T_i) = \text{ValidationTS}(T_i)$
- Validation-based protocol has been found to give greater degree of concurrency than locking/TSO if probability of conflicts is low.



Validation Test for Transaction T_j

- If for all T_i with $TS(T_i) < TS(T_j)$ either one of the following condition holds:
 - **finishTS(T_i) < startTS(T_j)**
 - **startTS(T_j) < finishTS(T_i) < validationTS(T_j)** and the set of data items written by T_i does not intersect with the set of data items read by T_j .

then validation succeeds and T_j can be committed.

- Otherwise, validation fails and T_j is aborted.
- Justification:
 - First condition applies when execution is not concurrent
 - The writes of T_j do not affect reads of T_i since they occur after T_i has finished its reads.
 - If the second condition holds, execution is concurrent, T_j does not read any item written by T_i .



Schedule Produced by Validation

- Example of schedule produced using validation

T_{25}	T_{26}
read(B)	read(B) $B := B - 50$ read(A) $A := A + 50$
read(A) <validate> display($A + B$)	<validate> write(B) write(A)



Multiversion Concurrency Control



Multiversion Schemes

- Multiversion schemes keep old versions of data item to increase concurrency. Several variants:
 - **Multiversion Timestamp Ordering**
 - **Multiversion Two-Phase Locking**
 - **Snapshot isolation**
- Key ideas:
 - Each successful **write** results in the creation of a new version of the data item written.
 - Use timestamps to label versions.
 - When a **read**(Q) operation is issued, select an appropriate version of Q based on the timestamp of the transaction issuing the read request, and return the value of the selected version.
- **reads** never have to wait as an appropriate version is returned immediately.



Multiversion Timestamp Ordering

- Each data item Q has a sequence of versions $\langle Q_1, Q_2, \dots, Q_m \rangle$. Each version Q_k contains three data fields:
 - **Content** -- the value of version Q_k .
 - **W-timestamp**(Q_k) -- timestamp of the transaction that created (wrote) version Q_k
 - **R-timestamp**(Q_k) -- largest timestamp of a transaction that successfully read version Q_k



Multiversion Timestamp Ordering (Cont)

- Suppose that transaction T_i issues a **read**(Q) or **write**(Q) operation. Let Q_k denote the version of Q whose write timestamp is the largest write timestamp less than or equal to $TS(T_i)$.
 1. If transaction T_i issues a **read**(Q), then
 - the value returned is the content of version Q_k
 - If $R\text{-timestamp}(Q_k) < TS(T_i)$, set $R\text{-timestamp}(Q_k) = TS(T_i)$,
 2. If transaction T_i issues a **write**(Q)
 1. if $TS(T_i) < R\text{-timestamp}(Q_k)$, then transaction T_i is rolled back.
 2. if $TS(T_i) = W\text{-timestamp}(Q_k)$, the contents of Q_k are overwritten
 3. Otherwise, a new version Q_i of Q is created
 - $W\text{-timestamp}(Q_i)$ and $R\text{-timestamp}(Q_i)$ are initialized to $TS(T_i)$.



Multiversion Timestamp Ordering (Cont)

- Observations
 - Reads always succeed
 - A write by T_i is rejected if some other transaction T_j that (in the serialization order defined by the timestamp values) should read T_i 's write, has already read a version created by a transaction older than T_i .
- Protocol guarantees serializability



Multiversion Two-Phase Locking

- Differentiates between read-only transactions and update transactions
- **Update transactions** acquire read and write locks, and hold all locks up to the end of the transaction. That is, update transactions follow rigorous two-phase locking.
 - Read of a data item returns the latest version of the item
 - The first **write** of Q by T_i results in the creation of a new version Q_i of the data item Q written
 - $W\text{-timestamp}(Q_i)$ set to ∞ initially
 - When update transaction T_i completes, commit processing occurs:
 - Value **ts-counter** stored in the database is used to assign timestamps
 - **ts-counter** is locked in two-phase manner
 - Set $TS(T_i) = \text{ts-counter} + 1$
 - Set $W\text{-timestamp}(Q_i) = TS(T_i)$ for all versions Q_i that it creates
 - **ts-counter** = **ts-counter** + 1



Multiversion Two-Phase Locking (Cont.)

- **Read-only transactions**
 - are assigned a timestamp = **ts-counter** when they start execution
 - follow the multiversion timestamp-ordering protocol for performing reads
 - Do not obtain any locks
- Read-only transactions that start after T_i increments **ts-counter** will see the values updated by T_i .
- Read-only transactions that start before T_i increments the **ts-counter** will see the value before the updates by T_i .
- Only serializable schedules are produced.



MVCC: Implementation Issues

- Creation of multiple versions increases storage overhead
 - Extra tuples
 - Extra space in each tuple for storing version information
- Versions can, however, be garbage collected
 - E.g., if Q has two versions Q5 and Q9, and the oldest active transaction has timestamp > 9 , then Q5 will never be required again
- Issues with
 - primary key and foreign key constraint checking
 - Indexing of records with multiple versions

See textbook for details



Snapshot Isolation

- Motivation: Decision support queries that read large amounts of data have concurrency conflicts with OLTP transactions that update a few rows
 - Poor performance results
- Solution 1: Use multiversion 2-phase locking
 - Give logical “snapshot” of database state to read only transaction
 - Reads performed on snapshot
 - Update (read-write) transactions use normal locking
 - Works well, but how does system know a transaction is read only?
- Solution 2 (partial): Give snapshot of database state to every transaction
 - Reads performed on snapshot
 - Use 2-phase locking on updated data items
 - Problem: variety of anomalies such as lost update can result
 - Better solution: snapshot isolation level (next slide)



Snapshot Isolation

- A transaction T1 executing with Snapshot Isolation
 - Takes snapshot of committed data at start
 - Always reads/modifies data in its own snapshot
 - Updates of concurrent transactions are not visible to T1
 - Writes of T1 complete when it commits
 - **First-committer-wins rule:**
 - 4 Commits only if no other concurrent transaction has already written data that T1 intends to write.

T1	T2	T3
W(Y := 1) Commit		
	Start R(X) □ 0 R(Y) □ 1	
		W(X:=2) W(Z:=3) Commit
	R(Z) □ 0 R(Y) □ 1 W(X:=3) Commit-Req Abort	

Concurrent updates not visible
 Own updates are visible
 Not first-committer of X
 Serialization error, T2 is rolled back



Snapshot Read

- Concurrent updates invisible to snapshot read

$X_0 = 100, Y_0 = 0$

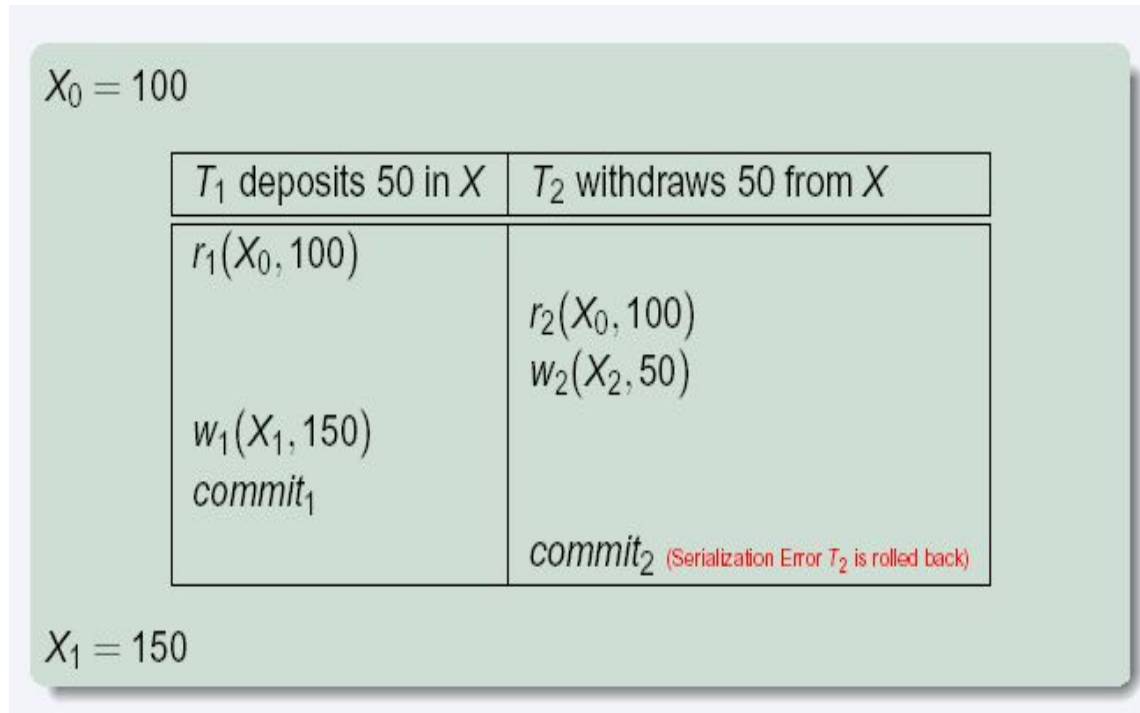
T_1 deposits 50 in Y	T_2 withdraws 50 from X
$r_1(X_0, 100)$ $r_1(Y_0, 0)$ $w_1(Y_1, 50)$ $r_1(X_0, 100)$ (update by T_2 not seen) $r_1(Y_1, 50)$ (can see its own updates)	$r_2(Y_0, 0)$ $r_2(X_0, 100)$ $w_2(X_2, 50)$ $r_2(Y_0, 0)$ (update by T_1 not seen)

$X_2 = 50, Y_1 = 50$

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Snapshot Write: First Committer Wins



- Variant: “**First-updater-wins**”
 - Check for concurrent updates when write occurs by locking item
 - 4 But lock should be held till all concurrent transactions have finished
 - (Oracle uses this plus some extra features)
 - Differs only in when abort occurs, otherwise equivalent



Benefits of SI

- Reads are *never* blocked,
 - and also don't block other txns activities
- Performance similar to Read Committed
- Avoids several anomalies
 - No dirty read, i.e. no read of uncommitted data
 - No lost update
 - I.e., update made by a transaction is overwritten by another transaction that did not see the update)
 - No non-repeatable read
 - I.e., if read is executed again, it will see the same value
- Problems with SI
 - SI does not always give serializable executions
 - Serializable: among two concurrent txns, one sees the effects of the other
 - In SI: neither sees the effects of the other
 - Result: Integrity constraints can be violated



Snapshot Isolation

- Example of problem with SI
 - Initially $A = 3$ and $B = 17$
 - Serial execution: $A = ??$, $B = ??$
 - if both transactions start at the same time, with snapshot isolation: $A = ??$, $B = ??$
- Called **skew write**
- Skew also occurs with inserts
 - E.g:
 - Find max order number among all orders
 - Create a new order with order number = previous max + 1
 - Two transaction can both create order with same number
 - Is an example of phantom phenomenon

T_i	T_j
read(A)	read(A) read(B)
read(B)	
$A=B$	$B=A$ write(B)
write(A)	



Snapshot Isolation Anomalies

- SI breaks serializability when transactions modify *different* items, each based on a previous state of the item the other modified
 - Not very common in practice
 - E.g., the TPC-C benchmark runs correctly under SI
 - when txns conflict due to modifying different data, there is usually also a shared item they both modify, so SI will abort one of them
 - But problems do occur
 - Application developers should be careful about write skew
- SI can also cause a read-only transaction anomaly, where read-only transaction may see an inconsistent state even if updaters are serializable
 - We omit details
- Using snapshots to verify primary/foreign key integrity can lead to inconsistency
 - Integrity constraint checking usually done outside of snapshot



Serializable Snapshot Isolation

- **Serializable snapshot isolation (SSI)**: extension of snapshot isolation that ensures serializability
- Snapshot isolation tracks write-write conflicts, but does not track read-write conflicts
 - Where T_i writes a data item Q , T_j reads an earlier version of Q , but T_j is serialized after T_i
- Idea: track read-write dependencies separately, and roll-back transactions where cycles can occur
 - Ensures serializability
 - Details in book
- Implemented in PostgreSQL from version 9.1 onwards
 - PostgreSQL implementation of SSI also uses index locking to detect phantom conflicts, thus ensuring true serializability



SI Implementations

- Snapshot isolation supported by many databases
 - Including Oracle, PostgreSQL, SQL Server, IBM DB2, etc
 - Isolation level can be set to snapshot isolation
- Oracle implements “first updater wins” rule (variant of “first committer wins”)
 - Concurrent writer check is done at time of write, not at commit time
 - Allows transactions to be rolled back earlier
- **Warning:** *even if isolation level is set to serializable, Oracle actually uses snapshot isolation*
 - Old versions of PostgreSQL prior to 9.1 did this too
 - Oracle and PostgreSQL < 9.1 do not support true serializable execution



Working Around SI Anomalies

- Can work around SI anomalies for specific queries by using **select .. for update** (supported e.g. in Oracle)
 - Example
 - **select max(orderno) from orders for update**
 - read value into local variable maxorder
 - insert into orders (maxorder+1, ...)
- **select for update (SFU) clause** treats all data read by the query as if it were also updated, preventing concurrent updates
- Can be added to queries to ensure serializability in many applications
 - Does not handle phantom phenomenon/predicate reads though



Weak Levels of Concurrency



Weak Levels of Consistency

- **Degree-two consistency:** differs from two-phase locking in that S-locks may be released at any time, and locks may be acquired at any time
 - X-locks must be held till end of transaction
 - Serializability is not guaranteed, programmer must ensure that no erroneous database state will occur]
- **Cursor stability:**
 - For reads, each tuple is locked, read, and lock is immediately released
 - X-locks are held till end of transaction
 - Special case of degree-two consistency



Weak Levels of Consistency in SQL

- SQL allows non-serializable executions
 - **Serializable**: is the default
 - **Repeatable read**: allows only committed records to be read, and repeating a read should return the same value (so read locks should be retained)
 - However, the phantom phenomenon need not be prevented
 - T1 may see some records inserted by T2, but may not see others inserted by T2
 - **Read committed**: same as degree two consistency, but most systems implement it as cursor-stability
 - **Read uncommitted**: allows even uncommitted data to be read
- In most database systems, read committed is the default consistency level
 - Can be changed as database configuration parameter, or per transaction
 - **set isolation level serializable**



Concurrency Control across User Interactions

- Many applications need transaction support across user interactions
 - Can't use locking for long durations
- Application level concurrency control
 - Each tuple has a version number
 - Transaction notes version number when reading tuple
 - **select** r.balance, r.version **into** :A, :version
from r **where** acctId = 23
 - When writing tuple, check that current version number is same as the version when tuple was read
 - **update** r **set** r.balance = r.balance + :deposit, r.version = r.version + 1
where acctId = 23 **and** r.version = :version



Concurrency Control across User Interactions

- Equivalent to **optimistic concurrency control without validating read set**
 - Unlike SI, reads are not guaranteed to be from a single snapshot.
 - Does not guarantee serializability
 - But avoids some anomalies such as “lost update anomaly”
- Used internally in Hibernate ORM system
- Implemented manually in many applications
- Version numbers stored in tuples can also be used to support first committer wins check of snapshot isolation



Advanced topics in Concurrency Control



Online Index Creation

- Problem: how to create an index on a large relation without affecting concurrent updates
 - Index construction may take a long time
 - Two-phase locking will block all concurrent updates
- Key ideas:
 - Build index on a snapshot of the relation, but keep track of all updates that occur after snapshot
 - Updates are not applied on the index at this point
 - Then apply subsequent updates to catch up
 - Acquire relation lock towards end of catchup phase to block concurrent updates
 - Catch up with remaining updates, and add index to system catalog
 - Subsequent transactions will find the index in catalog and update it



Concurrency in Index Structures

- Indices are unlike other database items in that their only job is to help in accessing data.
- Index-structures are typically accessed very often, much more than other database items.
 - Treating index-structures like other database items, e.g. by 2-phase locking of index nodes can lead to low concurrency.
- There are several index concurrency protocols where locks on internal nodes are released early, and not in a two-phase fashion.
 - It is acceptable to have nonserializable concurrent access to an index as long as the accuracy of the index is maintained.
 - In particular, the exact values read in an internal node of a B^+ -tree are irrelevant so long as we land up in the correct leaf node.



Concurrency in Index Structures (Cont.)

- **Crabbing protocol** used instead of two-phase locking on the nodes of the B^+ -tree during search/insertion/deletion:
 - First lock the root node in shared mode.
 - After locking all required children of a node in shared mode, release the lock on the node
 - During insertion/deletion, upgrade leaf node locks to exclusive mode.
 - When splitting or coalescing requires changes to a parent, lock the parent in exclusive mode.
- Above protocol can cause excessive deadlocks
 - Searches coming down the tree deadlock with updates going up the tree
 - Can abort and restart search, without affecting transaction
- The **B-link tree locking protocol** improves concurrency
 - Intuition: release lock on parent before acquiring lock on child
 - And deal with changes that may have happened between lock release and acquire



Concurrency Control in Main-Memory Databases

- Index locking protocols can be simplified with main-memory databases
 - Short term lock can be obtained on entire index for duration of an operation, serializing updates on the index
 - Avoids overheads of multiple lock acquire/release
 - No major penalty since operations finish fast, since there is no disk wait
- Latch-free techniques for data-structure update can speed up operations further



Latch-Free Data-structure Updates

- This code is not safe without latches if executed concurrently:

```
insert(value, head) {  
    node = new node  
    node->value = value  
    node->next = head  
    head = node  
}
```

- This code is safe

```
insert latchfree(head, value) {  
    node = new node  
    node->value = value  
    repeat  
        oldhead = head  
        node->next = oldhead  
        result = CAS(head, oldhead, node)  
    until (result == success)  
}
```



Latch-Free Data-structure Updates

- This code is not safe without latches if executed concurrently:

```
insert(value, head) {  
    node = new node  
    node->value = value  
    node->next = head  
    head = node  
}
```

- This code is safe

```
insert latchfree(head, value) {  
    node = new node  
    node->value = value  
    repeat  
        oldhead = head  
        node->next = oldhead  
        result = CAS(head, oldhead, node)  
    until (result == success)  
}
```



Latch-Free Data-structures (Cont.)

- Consider:

```
delete latchfree(head) {  
    /* This function is not quite safe; see explanation in text. */  
    repeat  
        oldhead = head  
        newhead = oldhead->next  
        result = CAS(head, oldhead, newhead)  
    until (result == success)  
}
```

- Above code is almost correct, but has a concurrency bug
 - P1 initiates delete with N1 as head; concurrently P2 deletes N1 and next node N2, and then reinserts N1 as head, with N3 as next
 - P1 may set head as N2 instead of N3.
- Known as ABA problem
- See book for details of how to avoid this problem



Concurrency Control with Operations

- Consider this non-two phase schedule, which preserves database integrity constraints
- Can be understood as transaction performing increment operation
 - E.g., $\text{increment}(A, -50)$, $\text{increment}(B, 50)$
 - As long as increment operation does not return actual value, increments can be reordered
 - *Increments commute*
 - New increment-mode lock to support reordering
 - Conflict matrix with increment lock mode
 - *Two increment operations do not conflict with each other*

T_1	T_2
$\text{read}(A)$ $A := A - 50$ $\text{write}(A)$	$\text{read}(B)$ $B := B - 10$ $\text{write}(B)$
$\text{read}(B)$ $B := B + 50$ $\text{write}(B)$	$\text{read}(A)$ $A := A + 10$ $\text{write}(A)$

	S	X	I
S	true	false	false
X	false	false	false
I	false	false	true



Concurrency Control with Operations (Cont.)

- Undo of $\text{increment}(v, n)$ is performed by $\text{increment}(v, -n)$
- $\text{Increment_conditional}(v, n)$:
 - Updates v by adding n to it, as long as final $v > 0$, fails otherwise
 - Can be used to model, e.g. number of available tickets, *avail_tickets*, for a concert
 - $\text{Increment_conditional}$ is NOT commutative
 - E.g., last few tickets for a concert
 - But reordering may still be acceptable



Real-Time Transaction Systems

- Transactions in a system may have deadlines within which they must be completed.
 - Hard deadline: missing deadline is an error
 - Firm deadline: value of transaction is 0 in case deadline is missed
 - Soft deadline: transaction still has some value if done after deadline
- Locking can cause blocking
- Optimistic concurrency control (validation protocol) has been shown to do well in a real-time setting



End of Chapter 18